

KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

Telephone: 080 – 23341652, 23348848 || Email: spp@kscst.org.in
 Website: www.kscst.org.in/spp.html or https://kscst.karnataka.gov.in/en

FORMAT FOR STUDENT PROJECT PROPOSAL FOR THE 49th SERIES OF STUDENT PROJECT PROGRAMME

(Handwritten proposals will not be accepted, please fill all the details in this MS word file, insert images / diagrams wherever necessary. Convert to pdf file, get it approved from the project guide / head of the department and principal of your institution. Keep ready the scanned pdf file of 1) Declaration and Endorsement 2) details of processing fees made and fill- up the Google Form.

<https://forms.gle/IXIR81MyoLWn871x7>

1.	Name of the College: Vivekananda Institute of Technology
2.	Project Title: PrintChakra: AI Powered Smart Print & Capture Solution
3.	Branch: Artificial Intelligence and Machine Learning
4.	Theme (as per KSCST poster): (The project proposals shall mandatorily be from one of the broad themes / areas. Visit website www.kscst.org.in/spp.html) Data Science, Cyber Security, Computing, Pattern Recognition and Image Processing, Signal Processing
5.	Name(s) of project guide(s): 1. Name: Dr. Shaila K Email id: dr.shailak.aiml@gmail.com Contact No.: 9964579958
6.	Name of Team Members (Strictly not more than four students in a batch): <i>(Type names in Capital Letters as provided in your college)</i> <i>(Please paste the latest passport size photograph adjacent to your respective names)</i>



Name: CHAMAN S
USN No.: 1VK22AI007
Email id: chamans79952@gmail.com
Mobile No: 6361005641



Name: GOWTHAM S
USN No.: 1VK22AI013
Email id: gr231202@gmail.com
Mobile No.: 8431792673



Name: KARTHIK KUMAR V GUPTA
USN No.: 1VK22AI016
Email id: guptakarthik41@gmail.com
Mobile No.: 7892096813

7. Team Leader of the Project:

Name: Chaman S
USN No.: 1VK22AI007
Email id: chamans79952@gmail.com
Mobile No.: 6361005641

8. Processing Fee Details (Through Online Payment only):
(processing fee of Rs. 1180/-)

- 1. UTR NO:** 543735294752
- 2. TRANSACTION ID:** T2512151442231472320512
- 3. NAME OF THE SENDER:** CHAMAN S
- 4. CONTACT NUMBER:** 6361005641
- 5. NAME OF THE BANK:** INDIAN BANK
- 6. PROCESSING FEES:** Rs. 1,180/-
- 7. DATE OF PAYMENT MADE:** 15/12/2025
- 8. TIME:** 2:42 PM
- 9. MODE OF PAYMENT MADE:** UPI

9. Date of commencement of the Project:
05/March/2025

10. Probable date of completion of the project:
10/ December/2025

Timeline Structure

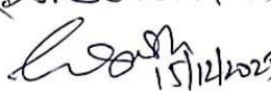
Task	Start Date	End Date	Duration	Milestone achievable
Project Initiation	4, March 2025	24, November 2025	8 months	Project topic finalized (PrintChakra), guide approval obtained, initial planning completed
Research Phase	4, March 2025	27, April 2025	2 months	Literature survey completed, problem statement defined, objectives finalized

		Development Phase	8, August 2025	17, October 2025	3 months	System design completed, application modules developed, integration of printer, OCR, LLM & Web application.
		Testing Phase	31, October 2025	28, November 2025	2 Months	Functional testing completed, bugs resolved, synchronization and performance validated
		Submission of PPT and Abstract to KSCST for conducting mid-term evaluation (Upload to Google Form)	Middle of March – 2 nd Week of April 2026			
		Final Review	28, November 2025	28, November 2025		Internal evaluation completed, guide approval for final submission
		Project Submission	28 November 2025	28 November 2025		PPT and abstract successfully completed
		Submission of Project Completion Report to KSCST (Upload to Google Form)	April / May 2026			
11.	Scope / Objectives of the project: AI-Driven Workflow Automation: Automates printing, scanning, and document conversion end-to-end using an integrated LLM (smolllm:135B), minimizing manual intervention. Natural Language-Based Control: Allows users to operate the system through simple voice or text commands interpreted by the LLM for intelligent task execution. AI-Based Image Processing and OCR: Enhances document images and extracts accurate text using image processing and OCR for reliable digitization. Offline-First Intelligent Operation: Executes scanning, processing, and LLM-driven automation over a local network without requiring internet connectivity. Scalable and Adaptive Architecture: Provides a cost-effective, modular solution that can scale across institutions and adapt to future AI-driven enhancements.					

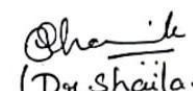
12.	Methodology: <pre> graph TD PC[PC] -- "1 Open" --> PUI(PC/Web Interface) PUI -- "2 Click on" --> SB(Scan Button) SB -- "3 Initializes Printer" --> P[Printer] P -- "4 Visuals" --> SM(Smartphone/Microprocessor) SM -- "5 Captures Image" --> SPP(Scanning & Processing) SPP -- "6 Stores in" --> SD[(Storage Device)] SD -- "Sent for" --> PP(Post Processing) PP -- "Organize/Edit" --> PUI PUI -- "WebSockets" --> SM PUI -- "USB" --> P SM -- "Initializes Scanner" --> SB </pre> PC/Web Interface Initialization: The user opens the PrintChakra web interface on the PC to configure print, scan, and document processing actions through manual controls or AI (LLM) commands. Printer Initialization and Document Preparation: The PC initializes the printer via USB/Wi-Fi and outputs the document, preparing it for capture while system parameters are set. Scanner Activation via Mobile/Device: On user or LLM instruction, the smartphone/mobile device initializes the scanner and captures the printed document using the camera. Image Capture and Visual Transfer: The captured document image is transferred over the local network to the processing system for further analysis. Scanning, Image Processing, and OCR: The system performs image enhancement, edge detection, and OCR to extract structured text and digitize the document. Storage and Post-Processing: The processed data is stored securely in the storage device (MongoDB/GridFS) and sent for post-processing, where documents can be organized, edited, converted, or printed automatically using the AI-based LLM.
13.	Expected Outcome of the project: Intelligent Document Automation: Real-time AI-driven automation enables seamless printing, scanning, and document conversion with minimal human intervention. Accurate and Reliable Digitization: Image processing and OCR ensure high accuracy in text extraction and document enhancement, reducing manual errors. Accessible and User-Friendly Interaction: Natural language control via the LLM allows users to manage document workflows easily through voice or text commands. Efficient and Secure Offline Operation: Local network processing with secure storage improves reliability, data privacy, & overall system efficiency.

14.	Is the project proposed relevant to the Industry / Society or Institution? Yes / No: Yes, relevant to the society and institutions. If yes, please provide details of the industry / institution and contact details: The project is intended for educational institutions, government offices, examination departments, libraries, and administrative organizations that manage high volumes of printed and scanned documents. It provides an AI-powered document automation and management system for printing, scanning, OCR-based digitization, and intelligent document conversion, particularly suitable for offline or secure institutional environments. Contact details: Institutional-level deployment (college offices, government departments, and administrative units); no single external industry partner at the proposal stage. Industry support (technology / funds / use of final product): • Technology support: Provided through institutional infrastructure, academic mentorship, and open-source AI/OCR frameworks. • Use of final product: The final system can be directly used and deployed by institutions for document processing and automation. • Funds: No direct industry funding; project developed as an institution-focused and socially relevant solution. (Note: Preference will be given to those projects relevant to the industry / institution. Hence be specific in giving detailed information). Is the industry extending support - technology / funds / use the final product, please specify.)												
15.	Can the product or process to be developed in the project be taken up for filing a Patent? Yes / No: yes Prior Art search done? Yes/No: yes Note: If your answer is “Yes”, you may contact Patent Information Centre of KSCST. For more details, email: pic@kscst.org.in												
16.	Budget details (break-up details should be given): Note: KSCST will provide nominal grant support for carrying out the project by students if selected by the project selection committee. <table border="1" data-bbox="317 1444 1374 1776"> <thead> <tr> <th>Budget</th><th>Amount</th></tr> </thead> <tbody> <tr> <td>a) Materials / Consumables (Please specify)</td><td>3200.00</td></tr> <tr> <td>b) Labor (Describe)</td><td>200.00</td></tr> <tr> <td>c) Travel (Describe)</td><td>600.00</td></tr> <tr> <td>e) Miscellaneous (Please specify)</td><td>1000.00</td></tr> <tr> <td>Total</td><td>5000.00</td></tr> </tbody> </table>	Budget	Amount	a) Materials / Consumables (Please specify)	3200.00	b) Labor (Describe)	200.00	c) Travel (Describe)	600.00	e) Miscellaneous (Please specify)	1000.00	Total	5000.00
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e) Miscellaneous (Please specify)	1000.00												
Total	5000.00												
17.	Any other technical details (Please specify): Unified web-based interface accessible from both Android devices and laptops, integrated with OpenCV-based scanning, Tesseract OCR, Flask REST APIs, MongoDB with GridFS, JWT/SSL security, and LLM-based automation (smollm:135B) for intelligent printing, scanning, and document conversion.												

18.	SPP Coordinator (Identified by the college): Note: To be identified by the principal of the institution. The project proposals must be submitted to KSCST through SPP coordinator designated by the principal. Name: Dr. Suresh P M Email id: spm_dvg@yahoo.co.in Contact No.: 9886756991
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(Dr. Suresh P.M.)

 (Name & Signature of
 SPP Coordinator with Seal)
 Email id: spm_dvg@yahoo.co.in
 Contact No.: 9886756991




 (Dr. Shaila K.)
 Name and signature of the
 Project Guide with Seal:
 Email id: dr.shaila.k.aiml@gmail.com
 Contact No.: 9964579958
 Head of the Department
 Department of Artificial
 Intelligence & Machine Learning Eng.
 Vivekananda Institute of Technology
 Bangalore - 560 074.

DECLARATION

(From Project Students)

We, the project team hereby declare that the details enclosed in the project proposal (Title of the Project: PrintChakra: AI Powered Smart Print & Capture Solution, Branch: Artificial Intelligence and Machine Learning, College: Vivekananda Institute of Technology are true and correct to the best of our knowledge and belief. We undertake to inform the Karnataka State Council for Science and Technology (KSCST) of any changes to the project title or team members' names immediately through our project guide or the SPP Coordinator of our institution.

Additionally, we declare that the project work is original and not a result of copying or purchasing. We are committed to completing the project independently, with support from our faculty and project guide, while utilizing the facilities provided by the college. We will not engage in plagiarism and pledge to be sincere and dedicated in executing and completing the project as proposed.

We understand that if any of the above information is found to be false, untrue, or misleading, we may be held liable. We authorize the sharing of the project information contained in this proposal with KSCST, Bengaluru.

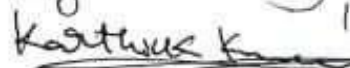
We acknowledge that the project team must exhibit and demonstrate the project, participate in the mid-term evaluation of sanctioned projects, and engage with experts. Additionally, we must exhibit the project at the Annual State-Level Poster Presentation and Exhibition, if selected. Should our team fail to participate in the mid-term evaluation or the Annual Exhibition (if selected), we understand that the supported project funds will be returned to KSCST.

We also enclose the endorsement form for KSCST, Bengaluru.

Name of the students with USN No.

1. Chaman S (1VK22AI007)
2. Gowtham S (1VK22AI013)
3. Karthik Kumar V Gupta (1VK22AI016)

Signature with date

 15/12/25
 15/12/25
 15/12/25

(Dr. S. Shailak. K.)

(Name & Signature of
SPP Coordinator with Seal)
Email id: spm_dvg@yahoo.co.in
Contact No.: 9886756991


(Dr. S. Shailak. K.)
Name and signature of the
Project Guide with Seal:
Email id: dr.shailak.aiml@gmail.com
Contact No.: 9964579958
Head of the Department
Department of Artificial
Intelligence & Machine Learning Eng.
Vivekananda Institute of Technology
Bengaluru - 560 074.



Janatha Education Society (Regd.)

Tel: 080-28437696

28437036

Fax: 080-28437944

VIVEKANANDA INSTITUTE OF TECHNOLOGY

(Approved by AICTE, New Delhi and Affiliated to V.T.U., Belgaum)

Gudimavu, Kumbalagodu Post, Kengeri Hobli, Bangalore - 560 074

Web: www.vitb.ac.in

E-mail: vkitprincipal@gmail.com

ENDORSEMENT

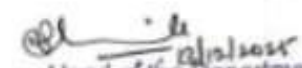
This is to certify that 1) Mr. Chaman S, 2) Mr. Gowtham S 3) Mr. Karthik Kumar V Gupta, are Bonafide student(s) of Department of Artificial Intelligence and Machine Learning, in the degree program of our institution. If the project proposal submitted by these students under the 49th series of Student Project Programme is selected by KSCST, we will provide the requisite laboratory / Computer / infrastructure support in our college / Institution. Further we also take necessary steps to see that the project team will exhibit / demonstrate their project in the mid-term evaluation of project and in the Annual State-Level Poster Presentation and Exhibition (if selected). If the student team fails to send the completed project report or fails to attend the evaluation in mid-term evaluation of sanctioned projects or fails to attend the Annual State-Level Poster Presentation and Exhibition (if selected), the supported project amount will be returned to KSCST.


Head of the Department
(Name & Signature of
Project Guide with Seal) Eng.
Vivekananda Institute of Technology
Email id: dr.shailak.aiml@gmail.com
Bangalore - 560 074.

Contact No.: 9964579958


(Name & Signature of
SPP Coordinator with Seal)
Email id: spm.dv@rahulco.in

Contact No.: 9886756991


Head of the Department
(Signature of HOD
with Seal) Department of Artificial
Intelligence & Machine Learning Eng.
Vivekananda Institute of Technol
Email id: dr.shailak.aiml@gmail.com
Bangalore - 560 074.

Contact No.: 9964579958


15/12/2024

(Signature of the Principal with Seal)

Email id: principal@vitb.ac.in

Contact No.: 9880373629

Principal

Vivekananda Institute of Technology
Gudimavu, Kumbalagodu Kengeri Hobli,
BANGALORE - 560074

DETAILS OF PROCESSING FEES MADE THROUGH NEFT / UPI PAYMENT

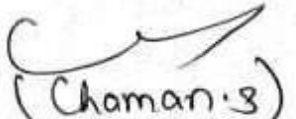
(Note: Include this page in the softcopy of the student project proposal. The student team shall furnish the details in the Google Form. It is informed to the students to 1) keep ready the softcopy of the project proposal and other documents and 2) Furnish the payment made details as processing fees and 3) update the details in the Google Form on the same day of payment made to KSCST by NEFT / UPI payment).

For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348848, email: finance@kscst.org.in

1. TITLE OF THE PROJECT	:	PrintChakra: AI Powered Smart Print & Capture Solution
2. NAME OF THE TEAM LEADER	:	Chaman S
3. EMAIL ID	:	chamans79952@gmail.com
4. CONTACT MOBILE NO.	:	6361005641

PAYMENT MADE DETAILS

5. BANK REF. NO. / UTR NO. / UPI No. (12 digits)	:	543735294752
6. TRANSACTION ID	:	T2512151442231472320512
7. NAME OF THE SENDER / ACCOUNT HOLDER and CONTACT NUMBER	:	CHAMAN.S 6361005641
8. NAME OF THE BANK	:	INDIAN BANK
9. PROCESSING FEES	:	Rs. 1,180/- (Inclusive of 18% GST)
10. DATE OF PAYMENT MADE	:	15/12/2025
11. TIME	:	2:42 PM
12. MODE OF PAYMENT MADE (NEFT / UPI, PLEASE SPECIFY)	:	UPI


 (Chaman S)
 (Name & Signature of the team leader)


 13/12/2025
 (Name & Signature of Project Guide / SPP Co-ordinator / HOD with Seal)
 Head of the Department
 Department of Artificial Intelligence & Machine Learning Eng.
 Vivekananda Institute of Technology
 Bangalore - 560 074.

KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

49th SERIES OF STUDENT PROJECT PROGRAMME (SPP)

(Note: This page is for information about bank details of KSCST to the student team and college / institution and not to include this page in the project proposal softcopy)

BANK ACCOUNT DETAILS OF KSCST

Name and address of the Institution	Karnataka State Council for Science and Technology, IISc Campus, Bangalore -560012
Account holder's name / Designation	Secretary, Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank, IISc Campus Branch, Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank, Indian Institute of Science, Bangalore-560012

BANK DETAILS

Name of the Agency	Karnataka State Council for Science and Technology IISc Campus, Bangalore - 560012
Account holder's name / Designation	Secretary , Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank IISc Campus Branch Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank Indian Institute of Science Bangalore-560012

Note: For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348848 during office hours between 3.00 pm to 5 pm, email: finance@kscst.org.in with a copy to spp@kscst.org.in



केनरा बैंक Canara Bank

भारत सरकार का उपक्रम

A Government of India Undertaking

सिंडिकेट Syndicate

KARNATAKA STATE COUNCIL FOR SCIENCE



0683201000024@cnrb

BHIM
BHARAT INTERFACE FOR MONEY

UPI
UNIFIED PAYMENTS INTERFACE



भारत 2023 INDIA

वसुधैव कुटुम्बकम्

ONE EARTH • ONE FAMILY • ONE FUTURE

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